

Audio-First AI: Overcoming the Transcription Bottleneck in Enterprise RAG

A RAG system is only as good as its text. Learn how to architect audio ingestion pipelines that eliminate hallucinated transcriptions.

Executive Summary:

Ingesting meeting recordings, earnings calls, or lectures into an enterprise RAG system requires transcription. However, advanced models like Whisper are highly susceptible to hallucinations when faced with poor acoustics or background noise. A single hallucinated transcription permanently poisons the vector database. This paper outlines an audio-first ingestion architecture that bounds memory usage and sanitizes transcriptions before vectorization.

1. The Engineering Challenge

During the ingestion of hundreds of hours of archival audio, we passed multi-hour files directly to MLX-Whisper. This resulted in two critical failures: 1) Unbounded memory usage that crashed the container, and 2) Severe hallucination on noisy files, where the model generated repeating foreign characters (e.g., 'arimarimarim...') instead of transcribing the background hiss. This garbage text was embedded and stored, destroying retrieval quality.

In high-stakes enterprise environments, this kind of failure is unacceptable. When building systems for Fortune 500 organizations, relying on basic prompt engineering or out-of-the-box vendor SDKs frequently masks underlying architectural fragility. The goal is to move beyond simple proofs-of-concept into deterministic, resilient data flow.

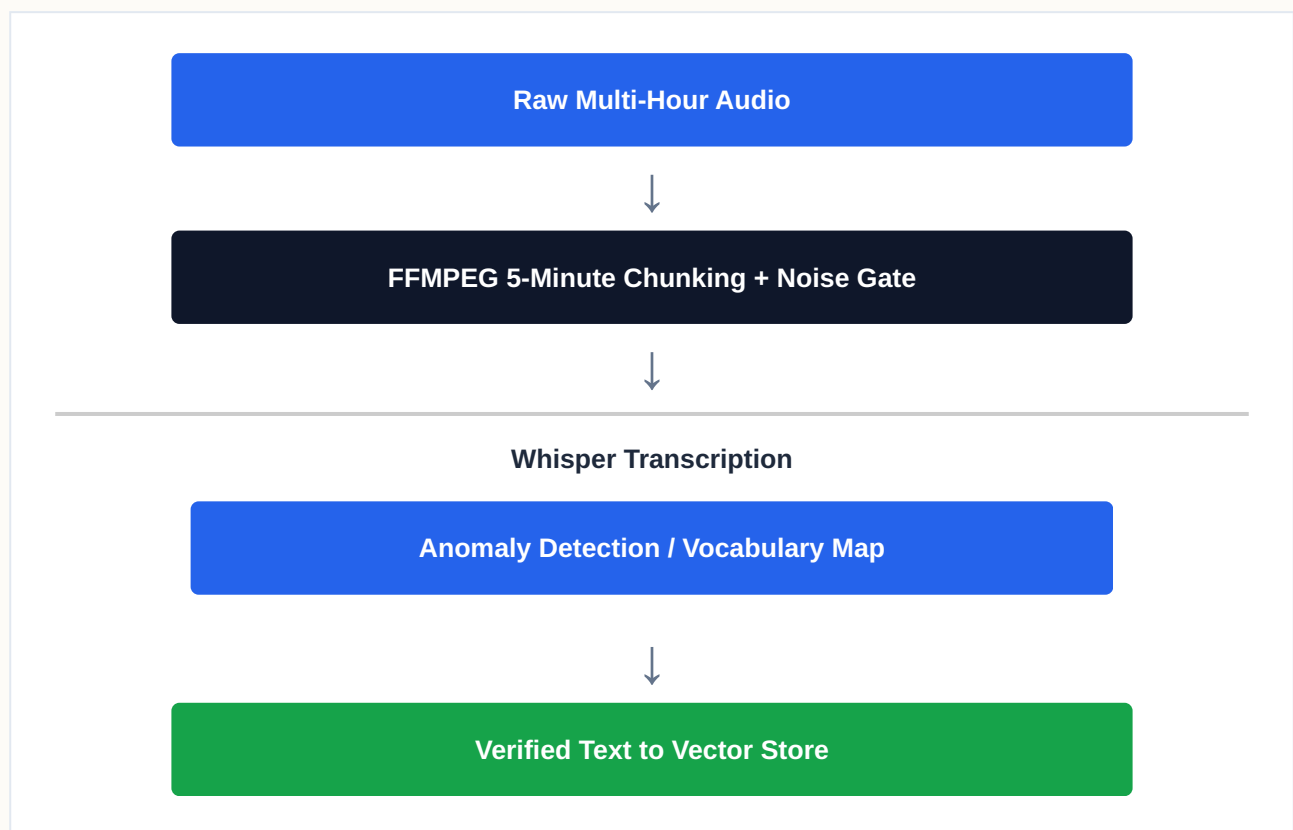
2. The Architectural Solution

We re-architected the ingestion pipeline. First, large audio files are chunked into 5-minute segments using `ffmpeg`, ensuring peak GPU memory is strictly bounded and providing granular progress reporting. Second, a mandatory pre-processing step uses acoustic filters to normalize volume and remove constant hiss. Finally, an anomaly detection gate runs on the output transcript, discarding repeating character sequences and enforcing rule-based

canonical vocabulary substitutions before the text is permitted to enter the ChromaDB vector store.

By enforcing this pattern at the architectural level, the system no longer relies on the "magic" of generative fluency. Instead, it relies on rigorous data engineering and precise, targeted models. This decoupling ensures that failures are isolated, auditable, and easily corrected without unwinding the entire data pipeline.

3. Structural Data Flow



4. Business Impact & ROI

For organizations operating at scale, migrating to this pattern fundamentally alters the cost and reliability curve. It shifts the burden of trust from the LLM's inherently probabilistic nature to a deterministic data engineering layer. The result is a system that can process massive throughput securely, drastically reducing API token burn and ensuring complete auditability for internal compliance boards.

